MIL-A- 50233 (AR)
7 July 1977
SUPERSEDING
See Section 6.

MILITARY SPECIFICATION

ADAPTER, WARHEAD SECTION: H4251

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of adapter entitled Adapter, Warhead Section: H4251 which is used with the Sprint Warhead Section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168	- Lot Numbering of Ammunition.

DRAWINGS

PICATINNY ARSENAL

8889030	- Adapter, Warhead Section: H4251.
8889031	- Ring, Aft Adapter.
8889032	- Pin, Pivoting.
9248927	- Screw.
9260286	- Box, Fiberboard, Packing, Ammunition for Adapter, Warhead (Aft): H4251.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

FSC 1190

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3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The adapter shall comply with all requirements specified on the drawing (dwg) 8889030, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Proof load. The adapter H4251 shall withstand a compressive force of 790 +20 -0 pounds applied for 6 + 1 minutes without damage. Testing shall be as described in 4.5.2.

3.6 Workmanship.

3.6.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.6.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.6.3 Burrs. Parts shall be free from burrs.

3.6.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.6.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 2 complete assemblies (see dwg 8889030) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Proof load (see 3.5)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

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4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

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4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Pin, pivoting (see dwg 8889032).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect small (body) diameter	Gage
102.	Incorrect body length "A" (see Tabulation)	Gage
103.	Incorrect total length "C" (see Tabulation)	Gage
104.	Incorrect large diameter "D" (see Tabulation)	Gage
105.	Incorrect total length "B" (see Tabulation)	Gage
106.	Evidence of poor workmanship (see 3.6)	Visual
107.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
201.	Protective finish improper, or damaged exposing base metal (see note on Dwg)	Visual
202.	Chamfer (4 places) missing, or improper	Visual

4.3.2.1.2 Ring, aft adapter (see dwg 8889031).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Mislocation of thru holes (13 holes) (Sec A-A)	Gage
102.	Incorrect diameter of thru holes (13 holes) (Sec A-A)	Gage
103.	Incorrect angularity of thru holes (13 holes) (Sec A-A)	Gage
104.	Mislocation of thru holes (4 holes) (Sec A-A)	Gage

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105.	Incorrect diameter of thru holes (4 holes) (Sec A-A)	Gage
106.	Incorrect diameter of Datum-D-top and bottom (Detail "A" and Sec A-A)	Gage
107.	Incorrect concentricity (Detail "A")	Gage
108.	Height of Datum -D- incorrect (Sec A-A)	Gage
109.	Incorrect diameter of thru holes (2 holes) (Sec C-C)	Gage
110.	Location of thru holes incorrect (2 holes) (Sec C-C)	Gage
111.	Distance from bottom of slots to bottom of datum -D- incorrect (2 places)	Gage
112.	Width of slots incorrect (2 places)	Gage
113.	Mislocation of slots (2 places)	Gage
114.	Incorrect outside diameter (Datum -B-)	Gage
115.	Incorrect length of notch	Gage
116.	Incorrect depth of notch	Gage
117.	Location of notch incorrect	Gage
118.	Radii at notch missing or improper (4 places)	Gage
119.	Incorrect diameter of intersecting holes at spot faced holes (2 holes)	Gage
120.	Mislocation of intersecting holes at spotfaced holes (2 holes)	Gage
121.	Objective evidence relating to material missing or improper (see 3.1)	Visual

Minor:

201.	Spotface incomplete or missing (Sec C-C)	Visual
202.	Evidence of poor workmanship (see 3.6)	Visual

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4.3.2.1.3 Adapter, Warhead Section: H4251 (see dwg 8889030).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Component damaged, improper, missing, or improperly assembled	Manual/Visual
102.	Rivet (26), or insert (4) extended above surface	Manual/Visual
103.	Areas described in note on dwg not free from protective fininh	Visual
104.	Evidence of poor workmanship (see 3.6)	Visual
105.	Incorrect pitch diameter of thread (MAX - 2 places)	Gage
106.	Incorrect minor diameter of thread (MAX - 2 places)	Gage
107.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
201.	Protective finish improper or damaged exposing base metal	Visual
202.	Marking missing, improper, illegible, or misplaced	Visual
203.	Identification plate missing, or is improperly positioned	Visual

4.3.2.1.4 Screw (see dwg 9248927).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Pitch diameter of thread (min.) incorrect	Gage
102.	Major diameter of thread (max.) incorrect	Gage
103.	Overall length of screw incorrect	Gage

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104. Angularity of screw head incorrect.....Gage
105. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor:
201. Evidence of poor workmanship (see 3.6).....Visual

4.3.2.1.5 Box, fiberboard, packing, ammunition for adapter, warhead (Aft): H4251 (see dwg 9260286).

Categories	Characteristics	Method of Inspection
Major:		
101.	Packing not in compliance with instructions on dwg	Visual
102.	Incorrect weight (Total)	Gage
103.	Marking, missing, incomplete, improper, illegible, or is improperly positioned	Visual
104.	Incorrect inside dimensions of box	Gage
105.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.6)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Proof load	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.4.3.1 Proof load. A sample shall be selected at random in accordance with Table IV. The sample shall be tested in accordance with 4.4.1. If any item fails to comply with 3.5, the lot shall be rejected.

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Table IV

Sample sizes for non-destructive testing

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major and Minor characteristics and tests. Inspection and testing shall be performed with the acceptance test equipment, operating instructions and calibration instructions prepared by the contractor. The contractor shall obtain approval of such designs, operating instructions and special calibration instructions as specified in 6.3 prior to use on the contract.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Proof load. The adapter shall be mounted to an approved fixture in the identical manner the warhead section (SPRINT) is attached. A compressive force or load of 790 +20 -0 pounds shall be applied for 6 + 1 minute. The weight of the force shall inpinge on the 13 bolts and 2 pinions supporting the load. An approved instrument to measure the applied force or the load shall be used. After 6 + 1 minute has elapsed, the adapter shall be dismounted from the fixture and examined for damage (see 6.4).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The adapter shall be packaged in accordance with dwg 9260286 using the specified moisture barrier.

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5.1.2 Level C. The adapter shall be packaged in accordance with dwg 9260286 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The adapter shall be packed in accordance with dwg 9260286.

5.3 Marking. Marking shall be as specified in accordance with dwg 9260286.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

6.4 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2244 dated 8 May 1972 and change dated 30 August 1972.

Custodian:
Army - AR

Preparing activity:
Army - AR

Project No. 1190-A102

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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